Site Generator Failure

Field Manual — Setup and Troubleshooting Reference

Use case definition. A site generator converts engine fuel energy into electrical power for field loads. Abnormal behavior includes failure to start, oil-pressure or coolant alarms, unstable voltage/frequency, overload or underload, breaker failure, fuel restriction, smoke, vibration, and trips during motor starting. Troubleshooting protects the engine, alternator, switchgear, and connected process equipment.

Manual setup and use. Use the engine and alternator manuals, protection settings, load-bank procedure, and OEM/site alarm or trip limits. Values are training/reference values.

Safety note. Treat fuel, exhaust, live electrical equipment, hot surfaces, and rotating parts as hazards. Only authorized personnel may test energized equipment.

Quick-use workflow

Use this sequence for every event: establish a safe operating state; capture the initial numerical readings; compare each value with the stated reference; apply the trigger-based action; correct the identified component or process path; restore operation gradually; and confirm the expected numerical result. If an OEM/site alarm or trip limit differs from a training value, the OEM/site limit governs.

Scenario 1: Fails to start

Field Condition

cranking 165 RPM; battery 21.4 V from 24–26; start attempts 3; fuel pressure 0.8 barg

Problem

Low battery voltage or fuel delivery is preventing start.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses battery below 23 V.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when battery 24–26 V and start at 1 attempt.

Expected Result

battery 24–26 V and start at 1 attempt

Scenario 2: Low oil pressure

Field Condition

oil pressure 1.2 barg at 1,500 RPM from 3.0–4.5; oil 82 °C

Problem

Low pressure indicates low level, filter restriction, pump issue, or sensor fault.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses pressure below 2.5 barg.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.

4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when pressure 3.0–4.5 barg.

Expected Result

pressure 3.0–4.5 barg

Scenario 3: High coolant temperature

Field Condition

coolant 104 °C from 75–90; load 78 %; radiator outlet airflow low

Problem

Cooling restriction or overload is causing overheating.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses temperature above 100 °C.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when coolant 75–90 °C.

Expected Result

coolant 75–90 °C

Scenario 4: Low coolant level

Field Condition

expansion tank 18 % from 45–70; coolant 88 °C; alarm active

Problem

Low coolant can cause rapid overheating and air ingress.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses level below 30 %.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when level 45–70 % after leak assessment.

Expected Result

level 45–70 % after leak assessment

Scenario 5: High exhaust temperature

Field Condition

cylinder 5 exhaust 612 °C from 430–520; load 64 %; spread 118 °C

Problem

Cylinder imbalance, injector, air, or exhaust restriction is likely.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses temperature above 580 °C or spread above 80 °C.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when all cylinders within approved spread and 430–520 °C.

Expected Result

all cylinders within approved spread and 430–520 °C

Scenario 6: Low battery voltage

Field Condition

battery 22.6 V during crank; charger output 0.4 A from 2–6; terminal corrosion visible

Problem

Charging failure or battery deterioration is preventing reliable starting.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses voltage below 23 V.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when battery 24–26 V and charger 2–6 A.

Expected Result

battery 24–26 V and charger 2–6 A

Scenario 7: Voltage instability

Field Condition

output 392–438 V; normal 400–415; load 58 %; AVR alarm

Problem

AVR, sensing, excitation, or load imbalance is causing voltage variation.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses swing above 10 V.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when output 400–415 V.

Expected Result

output 400–415 V

Scenario 8: Frequency instability

Field Condition

frequency 47.2–52.8 Hz; speed 1,416–1,584 RPM; load 62 %

Problem

Governor or fuel-control instability is producing speed oscillation.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses frequency outside 49–51 Hz.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when frequency 49–51 Hz and speed near 1,500 RPM.

Expected Result

frequency 49–51 Hz and speed near 1,500 RPM

Scenario 9: Overload

Field Condition

generator 880 kW at 96 % load; current 1,270 A from 1,050–1,180; coolant 97 °C

Problem

Connected load exceeds the stable operating range.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses load above 90 %.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when load 60–80 % after load shedding.

Expected Result

load 60–80 % after load shedding

Scenario 10: Underload

Field Condition

generator 42 kW at 4 % load for 6 h; exhaust 270 °C from 350–500

Problem

Extended low load can cause poor combustion and wet stacking.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses load below 15 % for extended operation.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.

4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when load 30–70 % or approved load-bank run.

Expected Result

load 30–70 % or approved load-bank run

Scenario 11: Breaker failure

Field Condition

generator 415 V; breaker command closed; bus 0 V; close coil 0 A

Problem

Breaker mechanism, control power, or interlock is preventing connection.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses bus remains 0 V after close command.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when breaker closed with 415 V bus and interlocks healthy.

Expected Result

breaker closed with 415 V bus and interlocks healthy

Scenario 12: Fuel restriction

Field Condition

fuel inlet vacuum −0.42 bar from −0.05 to −0.15; load 70 %; RPM 1,460

Problem

Blocked filter or supply line is starving the engine.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses vacuum below −0.25 bar.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when vacuum −0.05 to −0.15 bar and RPM 1,500.

Expected Result

vacuum −0.05 to −0.15 bar and RPM 1,500

Scenario 13: Excessive smoke

Field Condition

black smoke at 72 % load; exhaust 590 °C; turbo boost 0.8 barg from 1.2–1.6

Problem

Insufficient air, injector fault, or overload is causing incomplete combustion.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses boost below 1.0 barg or smoke persists >30 s.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when boost 1.2–1.6 barg and smoke clears.

Expected Result

boost 1.2–1.6 barg and smoke clears

Scenario 14: Abnormal vibration

Field Condition

engine vibration 9.6 mm/s RMS from 2–4; speed 1,500 RPM; bearing 86 °C

Problem

Loose mounts, misalignment, or rotating imbalance is likely.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses vibration above 7 mm/s RMS.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when vibration 2–4 mm/s RMS.

Expected Result

vibration 2–4 mm/s RMS

Scenario 15: Trip during motor starting

Field Condition

voltage falls 415 to 356 V; frequency 50 to 44 Hz; motor inrush 1,420 A; generator trips

Problem

Motor starting demand exceeds generator transient capability or start sequence is incorrect.

What Can Be Done

1. Confirm the abnormal value with the local indication and an independent calibrated reading; apply the approved unload, trip, or isolation action if it crosses voltage below 390 V or frequency below 48 Hz.
2. Compare the reading with the stated normal reference and check the associated process values for a consistent change.
3. Identify the likely cause by testing the named component or process path under the approved permit/LOTO conditions; do not open pressurized equipment.
4. Correct the specific restriction, setting, connection, lubrication, cooling, or mechanical defect identified in Step 3.
5. Restore operation in a controlled ramp and accept the equipment only when staged start with voltage above 400 V and frequency 49–51 Hz.

Expected Result

staged start with voltage above 400 V and frequency 49–51 Hz